Practical-11

Aim: To represent a real-world problem using propositional logic or first-order logic and apply logical inference techniques to derive valid conclusions using an AI reasoning approach.

CODE:

```
facts = {"raining"}
print("Initial facts:", facts)

rules = {
    "raining": "road_wet",
    "road_wet": "drive_carefully",
    "drive_carefully": "reduce_speed"
}


changed = True

while changed:
    changed = False

    for condition, conclusion in rules.items():
        if condition in facts and conclusion not in facts:
            facts.add(conclusion)
            changed = True

print("Facts after interface:")
for fact in facts:
  print(fact)
```

OUTPUT:

```
Facts after interface:
road_wet
reduce_speed
drive_carefully
raining
```


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Conclusion:

Thus, a real-world problem was successfully represented using propositional or first-order logic. Logical inference was performed to derive valid conclusions from the given facts and rules. This practical helped us understand how AI systems use logical reasoning to solve problems and make decisions.